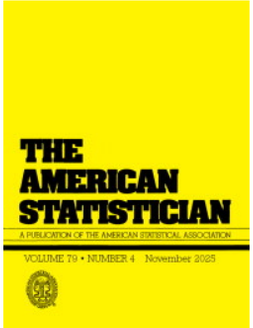

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Beyond the Yard Line: Accommodating Rounded Sports Data in Statistical Models

Amanda K. Glazer*, Layla Parast, and Mevin B. Hooten

Department of Statistics and Data Sciences, The University of Texas at Austin

*amanda.glazer@austin.utexas.edu

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Abstract

In American football, rushing, passing, and receiving yards are recorded as whole numbers, but using a unique method of rounding, even though the true yardage is continuous and recorded precisely on the field. This rounding introduces measurement error that is systematically ignored in most statistical analyses of these data. Beyond rounding, football yardage presents additional challenges: it can take on negative values and is strongly skewed. These characteristics complicate distributional assumptions and propagate rounding effects. We illustrate the consequences of these issues using data from running backs during the 2023 National Football League regular season. We show that appropriately modeling play-level yardage as a discrete, skewed, and possibly negative quantity, without access to the true values, is important to reconcile the approach with the data generation process. We compare candidate models that correctly incorporate rounding from a model checking and validation perspective. Our findings underscore the broader importance of accounting for discretization and asymmetry in sports analytics and other fields, where recorded data may mask the underlying measurement process in ways that meaningfully affect statistical conclusions.

Keywords: American football, yardage, rounding, measurement error, discrete data modeling, skewed distributions

1 Introduction

In American football (hereafter "football"), every yard counts, but not every yard is counted in the same way. While rushing, receiving, and passing yards are technically continuous-valued, they are recorded as whole numbers, rounded in a way that results in integer-valued data for each play. Figure 1 illustrates the current recording practice using four example offensive carries (i.e., movements of the ball).

In what follows, we focus on plays by running backs, players whose primary role is to carry the ball on running plays, called "rushes" or "carries," after receiving a handoff from the quarterback (as opposed to catching a forward pass). In Figure 1, each red segment reflects a play: The red dot marks where a carry *j* began (b_j) (i.e., the line of scrimmage; even though the player may actually gain possession of the ball behind the line of scrimmage), and the tip of the arrow shows where the player was downed (d_j). The length of the arrow represents the actual yards gained on the field ($\tilde{y}_j$), shown in the last column of the "truth" table in Figure 1. However, the official recorded yardage is rounded in a non-obvious way. Rather than rounding the true yardage to the nearest whole number directly, the recorded yards y_j are calculated as $y_j = \lfloor b_j \rfloor - \lfloor d_j \rfloor$, where $\lfloor \cdot \rfloor$ represents the floor function (i.e., rounding down; which, hereafter we refer to as "rounding"), as shown in the "recorded" table in Figure 1. It is important to note that *on the field* the ball is placed exactly where the player was downed; the rounding occurs only in the official recorded data. To our knowledge, there is no publicly available data set that captures the precise, unrounded yardage for each play. While precise *player* tracking has become more common, this is not the same as the yards gained on a play; we discuss this distinction in Section 4.

Why does this matter? Recorded yardage is a central component of player evaluation and is widely used in football analytics to measure and predict both individual and team performance across contexts. This includes fantasy sports, in-game decision-making, player contracts, trades, and betting markets. In many applications, yards gained serves either as the primary outcome in predictive modeling or as a key covariate in models estimating win probability or expected points. Despite this, the standard practice is to treat recorded yardage data as continuous-valued, overlooking the fact that they are rounded. Moreover, the recorded yardage is not simply a rounded down version of the true yardage (i.e., $\lfloor \tilde{y}_j \rfloor$). This is an important consideration when specifying valid generative models that formally recognize modeling choices for the true unrounded yardage.

Treating recorded yardage as continuous-valued is likely only a practical simplification in most studies that have done so (e.g., Biro and Walker, 2024); the fact that the resulting estimated parameters are not associated with the intended model for true yardage may be surprising to analysts. Thus, we show how to account for the unique style of rounding in football yardage data when constructing parametric statistical models for cases where the true response variable is not available. In doing so, our interest is in analyzing player performance with a generative model rather than modeling the rounded yardage as a surrogate for true yardage.

The issue of rounding is neither new, nor unique to football, nor is it limited to yardage statistics (Kowal et al., 2020). In football alone, several officially recorded measurements are

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routinely rounded, including field goal distances, time of possession (Carroll et al., 1988), player physical metrics (such as height, weight, and strength), 40-yard dash times, and passing or throwing distances. Rounding is similarly pervasive in other sports: minutes played in basketball, total passing and carrying distances in soccer, and hit distances in baseball are often reported as rounded values. More broadly, rounding arises across many domains. In education research, for instance, Likert scale responses in teaching evaluations are inherently discrete and often treated as continuous outcomes despite their ordinal nature (Boring et al., 2016). In health care quality measurement, star ratings for hospitals (e.g., the Centers for Medicare and Medicaid Services Star Ratings) are based on survey responses and performance metrics that are aggregated and rounded to whole number stars (Bilimoria and Barnard, 2021; CMS, 2025). In clinical trials, rounding continuous outcomes such as blood pressure or cholesterol levels can distort the interpretation of treatment effects, leading to inaccurate conclusions about the efficacy of medical interventions (Wen et al., 1993; Dong, 2015). Finally, in economic data reporting, rounding prices in the consumer price index can introduce minor inaccuracies in inflation statistics, impacting economic analysis and downstream policy decisions (Wilson and Abdirizak, 2017; Williams, 2006).

Even so, rounding of football yardage is different because of how it involves two rounding operations. But, beyond the unique style of rounding, football yardage presents a second modeling challenge: It can take on negative values. Unlike many other rounded variables discussed earlier, yards gained (or lost) on a play can be negative; for example, when a player is tackled behind the line of scrimmage. This feature underscores the importance of carefully considering distributional assumptions; models that assume only non-negative values may misrepresent the true data-generating process. A third feature of player performance data (e.g., yards gained) is that they are often skewed (to the right) due to the design of the game, offensive strategy, and biomechanics.

In what follows, we illustrate how to accommodate the aspects of rounding using data from running backs during the 2023 National Football League (NFL) regular season. Specifically, we provide a general framework to model true yards gained using a chosen base probability distribution while accounting for (a) the discrete and rounded nature of the recorded data, (b) the presence of negative and highly concentrated values, and (c) naturally occurring asymmetry. We begin with single-player modeling in Section 2, followed by a full multi-player analysis in Section 3. All code used to perform these analyses and generate the distributions is available at https://github.com/akglazer/rounded-sports-data.

2 Single-Player Yardage

On a given down in football, the offense can accumulate yards by rushing or passing the ball. For a given offensive player *i* who either receives or is handed the ball on carry *j*, we denote the true yards attained by that player as $\tilde{y}_{ij}$. Recall that the recorded yards are

$$y_{ij} = \lfloor b_{ij} \rfloor - \lfloor d_{ij} \rfloor, \quad (1)$$

where b_{ij} is the line of scrimmage and d_{ij} is point at which the player was downed. Both b_{ij} and d_{ij} are measured in yards from the opponent's end zone, ranging from 0 (at the goal line) to 100. [1] For example, Figure 2 shows the marginal yardage data for Bijan Robinson (of the Atlanta Falcons) in 2023.

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The integer values are somewhat unique because we cannot use classical discrete distributions such as the Poisson or negative binomial to model the data, due to negative values in y_{ij}. Even shifted versions of those distributions induce artificial constraints on the lower-bound for y_{ij} that do not allow them to serve as realistic generative models for the data.

We can, however, construct rounded distributions that are flexible using a transformation of variables. If we assume the true yardage is distributed as $\tilde{y}_{ij} \sim f_{\tilde{y}}(\tilde{y}_{ij} \mid \boldsymbol{\theta})$ for a chosen probability density function (PDF) $f_{\tilde{y}}$, then we seek the correct implied distribution for the rounded data y_{ij}. In what follows, we show that the transformation from $\tilde{y}_{ij}$ to y_{ij} is stochastic (given the available information) and can be expressed as $y_{ij} = \lfloor \tilde{y}_{ij} + \varepsilon_{ij} \rfloor$, where $0 < \varepsilon_{ij} < 1$.

To show this, we substitute $b_{ij} = \tilde{y}_{ij} + d_{ij}$ into (1) and then expand d_{ij} as $\lfloor d_{ij} \rfloor + \varepsilon_{ij}$ such that

$$y_{ij} = \lfloor b_{ij} \rfloor - \lfloor d_{ij} \rfloor \quad (2)$$

$$= \lfloor \tilde{y}_{ij} + d_{ij} \rfloor - \lfloor d_{ij} \rfloor \quad (3)$$

$$= \lfloor \tilde{y}_{ij} + \lfloor d_{ij} \rfloor + \varepsilon_{ij} \rfloor - \lfloor d_{ij} \rfloor \quad (4)$$

$$= \lfloor \tilde{y}_{ij} + \varepsilon_{ij} \rfloor + \lfloor d_{ij} \rfloor - \lfloor d_{ij} \rfloor \quad (5)$$

$$= \lfloor \tilde{y}_{ij} + \varepsilon_{ij} \rfloor, \quad (6)$$

where, in (5) we used the fact that $\lfloor d_{ij} \rfloor$ is an integer and does not affect the rounding of $\lfloor \tilde{y}_{ij} + \varepsilon_{ij} \rfloor$. We note that $\varepsilon_{ij} = d_{ij} - \lfloor d_{ij} \rfloor$ and, if the variance of d_{ij} is large relative to 1, then $\varepsilon_{ij} \xrightarrow{D} u_{ij}$ such that $u_{ij} \sim \text{Unif}(0,1)$. These assumptions are reasonably well-met in the football setting, thus we use $y_{ij} \approx \lfloor \tilde{y}_{ij} + u_{ij} \rfloor$ hereafter.

Dropping the subscripts for clarity, given the desired model for the true yardage $f_{\tilde{y}}$ and assuming $\tilde{y}$ and u are independent (which is reasonable in this case), we can derive the model for $\tilde{y}_u = \tilde{y} + u$ using a convolution

$$f_{\tilde{y}_u}(\tilde{y}_u) = \int_{-\infty}^{\infty} f_u(\tilde{y}_u - \tilde{y}) f_{\tilde{y}}(\tilde{y}) d\tilde{y}. \quad (7)$$

We show that (7) equates to the difference of two known cumulative distribution functions (CDFs) that depend on the chosen model for true yardage. To see this, we refine the limits of integration in (7) because the uniform PDF is $f_u(\tilde{y}_u - \tilde{y}) = 1$ for $0 < \tilde{y}_u - \tilde{y} < 1$. Thus,

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$$f_{\tilde{y}_u}(\tilde{y}_u) = \int_{-\infty}^{\infty} f_u(\tilde{y}_u - \tilde{y}) f_{\tilde{y}}(\tilde{y}) d\tilde{y} \quad (8)$$

$$= \int_{\tilde{y}_u - 1}^{\tilde{y}_u} f_{\tilde{y}}(\tilde{y}) d\tilde{y} \quad (9)$$

$$= F_{\tilde{y}}\left(\tilde{y}_u\right) - F_{\tilde{y}}\left(\tilde{y}_u - 1\right), \quad (10)$$

where, $F_{\tilde{y}}$ is the CDF associated with PDF $f_{\tilde{y}}$ and the form in (10) is familiar as a distribution associated with a rounded variable. However, an important distinction in this case is that $\tilde{y}_u$ are continuous-valued random variables (not rounded).

Rounding comes in the next step where we seek the distribution for the recorded yardage data y_{ij}. It is well-known that the correct probability mass function (PMF) for a rounded observation is a difference of CDFs associated with the unrounded variable (Davison, 2003). Thus, the correct PMF for recorded yardage data y_{ij} is

$$p_y(y_{ij}) = \int_{y_{ij}}^{y_{ij}+1} f_{\tilde{y}_u}(\tilde{y}_u) d\tilde{y}_u \quad (11)$$

$$= F_{\tilde{y}_u}\left(y_{ij} + 1\right) - F_{\tilde{y}_u}\left(y_{ij}\right), \quad (12)$$

where $F_{\tilde{y}_u}$ is the CDF for $\tilde{y}_u$ associated with $f_{\tilde{y}_u}$ from (10). Thus, if the base PDF $f_{\tilde{y}}$ and CDF $F_{\tilde{y}}$ are well-specified and the CDF $F_{\tilde{y}_u}$ can be derived, then the PMF p_y in (12) is available to use as a data model for y_{ij}.

Depending on the choice of the base distribution, the PMF p_y in (12) may not be available in closed-form. However, we show that p_y can be closely approximated in many cases using Gauss-Legendre quadrature which yields a fast and stable numerical calculation for the PMF. To see this, we express the integrand in (12) using (10), then change the variables so that the result can be written as a convolution involving a uniform which can be approximated using Gauss-Legendre quadrature. Thus, dropping the subscripts for clarity, we have

$$p_y(y) = \int_{y}^{y+1} f_{\tilde{y}_u}(v) dv \quad (13)$$

$$= \int_{y}^{y+1} F_{\tilde{y}}(v) - F_{\tilde{y}}(v-1) dv \quad (14)$$

$$= \int_{-1}^{1} \left(F_{\tilde{y}}\left(y + \frac{\theta}{2} + \frac{1}{2} \right) - F_{\tilde{y}}\left(y + \frac{\theta}{2} + \frac{1}{2} - 1 \right) \right) \frac{1}{2} d\theta \quad (15)$$

$$= \int_{-1}^{1} \left(F_{\tilde{y}}\left(y + \frac{\theta}{2} + \frac{1}{2} \right) - F_{\tilde{y}}\left(y + \frac{\theta}{2} - \frac{1}{2} \right) \right) f_{\theta}(\theta) d\theta, \quad (16)$$

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for the change of variables $v = y + \frac{\theta}{2} + \frac{1}{2}$, where f_θ is $\text{Uniform}(-1,1)$.

The integral in (16) can be computed numerically using Gauss-Legendre quadrature. In this case, the PMF for *y* can be approximated as

$$p_y(y) \approx \sum_{l=1}^{L} \frac{1}{(1-\theta_l^2)P_L'(\theta_l)^2}\left(F_{\tilde{y}}\left(y+\frac{\theta_l}{2}+\frac{1}{2}\right)-F_{\tilde{y}}\left(y+\frac{\theta_l}{2}-\frac{1}{2}\right)\right), \quad (17)$$

where θ_l are roots of the Legendre polynomial $P_L(\theta)$ of degree *L* (in our examples that follow, $L = 7$ provides a very accurate approximation). Thus, after we specify the distribution ($F_{\tilde{y}}$) for the true (unrounded) yardage, a PMF for the data can be computed using (17) to calculate the log likelihood as needed to fit the model using maximum likelihood or Bayesian approaches.

The advantages of using the correct rounded distribution as a model for y_{ij} are well studied (e.g., Davison, 2003). For example, the generative nature of the distribution can be used to obtain predictive realizations with the correct support. Furthermore, the resulting inference corresponds to the actual distribution specified for the true (unrounded) response variables. Davison (2003) discussed how rounded data lead to information lost when estimating parameters and that, when used in conjunction with mixture models, the resulting likelihood can be unstable due to strong multimodality effects.

2.1 Analysis of Single-Player Yardage Data

Using a Bayesian approach, we specified two models for the true yardage $\tilde{y}$, each based on a different underlying distribution: 1) a normal distribution with PDF

$$f_{\text{Norm}}(\tilde{y} \mid \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(\tilde{y}-\mu)^2}{2\sigma^2}\right), \quad (18)$$

and 2) an asymmetric Laplace distribution (ALD). We fit both models to the data shown in Figure 2. Following Williams et al. (2023), we describe the PDF for the ALD as

$$f_{\text{ALD}}(\tilde{y} \mid \mu, \sigma, \tau) = \frac{\tau(1-\tau)}{\sigma} \exp\left(-\frac{\tilde{y}-\mu}{\sigma}\left(\tau - 1_{\{\tilde{y}<\mu\}}\right)\right), \quad (19)$$

where τ controls the skewness (and $0 < \tau < 1$ in general, with $\tau < 0.5$ implying right-skew) and μ and σ control location and dispersion. The ALD is a skewed distribution for continuous-valued random variables that is commonly used in Bayesian quantile regression (e.g., Yang et al., 2016; Hooten and Hefley, 2019). Preliminary analyses with football yardage data revealed that the empirical distributions are often right-skewed and sharply peaked at μ as shown in Figure 2.

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We specified the correct PMFs as models for the recorded yardage data y_{ij} (i.e., $p_{\text{Norm}}(y_{ij} \mid \mu_{ij}, \sigma)$ and $p_{\text{ALD}}(y_{ij} \mid \mu_{ij}, \sigma, \tau)$) using (12) and modeled each location parameter as a linear function of covariates $\mathbf{x}_{ij}$ such that $\mu_{ij} = \mathbf{x}_{ij}^{'}\boldsymbol{\beta}$. We used vague priors for the parameters $\boldsymbol{\beta}$, σ, and τ:

$$\boldsymbol{\beta} \sim \text{N}(\mathbf{0}, 10^{12}\mathbf{I}), \quad (20)$$

$$\sigma^2 \sim \text{IG}(0.001, 1000), \quad (21)$$

$$\tau \sim \text{Beta}(1,1). \quad (22)$$

We used 10 covariates to predict yards gained: 1) "down," 2) "down squared," 3) "game seconds remaining," 4) "score differential," 5) "yards to go," 6) "line of scrimmage," 7) "days of rest before game," 8) "the opponent's rush expected points allowed (EPA)," 9) "homefield status," and 10) "travel distance to game." These covariates account for features at a range of time scales, from play-by-play (e.g., down and yards to go) up to the game-level (e.g., homefield status: home versus away). Due to the strong parity among teams in the NFL, we expect these covariates to be only weakly predictive, but there are clear mechanistic reasons for which we would expect certain relationships to be evident in the data; in Figure 3 we display recorded yardage versus each of these covariates for Bijan Robinson. For example, we expect a home field advantage (due to reduced crowd noise during offensive plays, familiarity with the stadium, etc.) which may result in greater yardage per carry when playing at home.

To demonstrate our approach, we analyzed data from the 2023 NFL season for Bijan Robinson (Figure 3). We checked the skewness and kurtosis assumptions of both models using posterior predictive *p*-values, and also compared them using DIC. During 2023, Bijan Robinson had 193 carries, approximately 10% of which resulted in negative recorded yardage. On the positive side, approximately 10% of carries resulted in gains of greater than 10 yards. We fit each model using Markov chain Monte Carlo (MCMC) based on 10^5 iterations, removing the first 20% as burn-in, and assessed convergence visually.

Using standardized covariates (except for homefield status which is binary), we fit the rounded-normal model first and checked the assumptions using posterior predictive *p*-values (Meng, 1994; Conn et al., 2018). Note that we used "down" as a quadratic variable instead of categorical because the latter resulted in worse predictive performance and algorithm instability due to the very small number of 4th downs in the dataset. As expected based on the skewness and heavy right tail evident in Figure 2, both *p*-values were near zero indicating lack of fit due to the inability of the normal to properly characterize the skewness and kurtosis in the data ($p_{\text{skew}} < 0.001$ and $p_{\text{kurt}} < 0.001$). By contrast, the posterior predictive *p*-values for the rounded-ALD model were both > 0.05 indicating less evidence for lack of fit ($p_{\text{skew}} = 0.06$ and $p_{\text{kurt}} = 0.09$). Similarly, from a predictive perspective, the deviance information criterion (DIC) for the rounded-ALD model was substantially better (i.e., lower) than that for the rounded-normal model ($\text{DIC}_{\text{Norm}} = 1247.6$ versus $\text{DIC}_{\text{ALD}} = 1134.6$).

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In terms of inference, Table 1 shows the posterior means, standard deviations, and 95% equal-tailed credible intervals for the regression coefficients ($\boldsymbol{\beta}$) resulting from fitting both models to the data.

Notable aspects of the results presented in Table 1 are 1) the effects are mostly weak, as expected, but 2) the posterior standard deviations are smaller by almost half for the rounded-ALD model; also 3) there were some sign changes between models in the posterior means, although those were in cases where the effects themselves were not strong, but 4) the third slope coefficient β_3 became prominent enough using the rounded-ALD model that we could consider its covariate (game seconds remaining) to be a helpful predictor of yardage for Bijan Robinson in 2023. In this case, the results suggest that when there was less time on the clock in the game, Bijan Robinson gained more yards per carry.

For completeness, the 95% equal-tailed credible interval for the dispersion parameter σ in the rounded-ALD model was $(0.88, 1.42)$. For the skewness parameter τ, the 95% credible interval was $(0.16, 0.29)$, which indicates substantial right-skewness.

Overall, the rounded model using the ALD as a base distribution for true yardage is superior to using the rounded-normal. The rounded-ALD model results in more precise estimates of the regression coefficients and allowed us to detect a helpful covariate (game seconds remaining) for predicting true yardage for a running back in the NFL. Furthermore, because we used the appropriately rounded version of the distribution as a model for the data, we know the model is generative for the recorded data we analyzed and were able to formally compare it to an alternative parametric generative model (i.e., rounded-normal), both from a model checking and model validation perspective.

3 Multi-Player Yardage

To extend the model to the multi-player setting (i.e., for all running backs in the NFL, including Bijan Robinson), we can use the same PMF for y_{ij} from (12) and generalize the model to account for variation in μ_{ij} across players. Thus, we specified a hierarchical rounded-ALD model using a random intercept for each player so that the model formulation is

$$y_{ij} \sim p_y(y_{ij} \mid \mu_{ij}, \sigma, \tau), \quad (23)$$

$$\mu_{ij} = \beta_{0,i} + \mathbf{x}_{ij}^{'}\boldsymbol{\beta}, \quad (24)$$

$$\beta_{0,i} \sim \mathrm{N}(\mu_0, \sigma_0^2), \quad (25)$$

$$\boldsymbol{\beta} \sim \mathrm{N}(\mathbf{0}, 10^{12}\mathbf{I}), \quad (26)$$

$$\mu_0 \sim \mathrm{N}(0, 1000), \quad (27)$$

$$\sigma_0^2 \sim \mathrm{IG}(0.001, 1000), \quad (28)$$

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$\sigma^2 \sim \text{IG}(0.001, 1000)$, (29)

$\tau \sim \text{Beta}(1,1)$, (30)

where, for the rounded-normal we omit the skewness parameter (τ) from the model statement. In this specification, we included 1) "age" and 2) "age squared," to the original set of 10 covariates: 3) "down," 4) "down squared," 5) "game seconds remaining," 6) "score differential," 7) "yards to go," 8) "line of scrimmage," 9) "days of rest before game," 10) "the opponent's rush expected points allowed (EPA)," 11) "homefield status," and 12) "travel distance to game" to yield 12 total. The age-related covariates allow us to account for other demographic differences among players in the multi-player models and we expected a quadratic effect where running backs may peak at some point in their career and then decline afterward (i.e., indicated by a positive effect for β_1 and a negative effect for β_2). All covariates except "homefield status" were standardized as described for the single-player model.

To fit the model, we used MCMC with Metropolis-Hastings updates for all parameters except μ_0 and σ_0^2 which had conjugate full-conditional distributions in both models. We used 20^5 MCMC iterations with a burn-in period of 20% of the iterations. We also compared the models using DIC, similar to the single-player analysis described previously. However, we used Gauss-Hermite quadrature to compute the deviance based on the integrated likelihood (i.e., marginalized over the random effects $\beta_{0,i}$). This was not necessary for the single-player models because they were not hierarchical so we used (12) to directly compute the deviance for those analyses.

The DIC for the rounded-normal model was $62,399.7$ and for the rounded-ALD model it was $57,780.5$. The effective number of parameters was approximately 14.9 for the rounded-normal and 16.0 for the rounded-ALD based on our model specifications. The smaller DIC implies the rounded-ALD model was superior to the rounded-normal, as we found in the single-player analyses. Thus, we use the rounded-ALD for final inference, but we compare the results to illustrate the differences in inference between the two models.

We fit our models to the available set of carries from running backs during the regular NFL season in 2023. The resulting dataset included $n = 133$ players combining for a total of $10,223$ carries. The number of carries per player varied from 1 to 250, thus the regularization induced by the partial pooling for the intercept $\beta_{0,i}$ allows for individual heterogeneity while also letting our inference for players with fewer carries borrow strength from those with more. Overall, the marginal posterior means and 95% equal-tailed credible intervals for each $\beta_{0,i}$ are shown in Figure 4.

We interpret the random intercepts as indicative of consistently better overall running backs while accounting for other factors that may affect the variability in yardage among them. For example, in Figure 4, we see that the 3 players with greatest posterior mean random effects were K. Williams, C. McCaffrey, and K. Mitchell for the rounded-normal model, but they were K. Williams, J. Cook, and D. Montgomery for the rounded-ALD. In fact, C. McCaffrey fell from 2nd to 17th in the ranked list when properly accounting for the skewness in the data using the ALD as the base distribution for true yardage. This occurs because he had a few

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unusually large yardage carries that the rounded-normal was sensitive to, but the rounded-ALD was not. The rounded-ALD model effectively discounted those extreme values for yardage in their influence on the regression because they were accommodated by the resulting right-skewed distribution.

In terms of inference on the fixed effects $\boldsymbol{\beta}$, shown in Table 2, we saw similar patterns as in the single-player analysis. As we found in the single-player analysis, the rounded-ALD resulted in smaller posterior standard deviations for the effects across all covariates which resulted in narrower 95% credible intervals as well. In fact, only 5 out of 12 effects had credible intervals that did not overlap zero in the rounded-normal analysis, but the rounded-ALD model resulted in 8 out of 12 significant effects. The 5 that were significant using the rounded-normal model were: 3) "down," 7) "yards to go," 8) "line of scrimmage," 10) "opp. EPA," and 11) "homefield status," whereas, using the rounded-ALD model, 3), 7), 8) and 10) remained significant, but 11) was not (barely). In addition, 1) "age," 4) "down squared" (nearly), 5) "game seconds remaining," and 6) "score differential" also did not overlap zero in the rounded-ALD results.

We interpret each of those significant effects associated with the rounded-ALD model in what follows. First, 1) "age" is positive, indicating a general increase in yardage with age of the running back, however, we note that there was a mild negative effect of 2) "age squared" which would indicate that players may have a peak performance period with age. We found a positive relationship between 3) down number and yardage and negative relationship with 4) down squared and yardage which provides evidence for a quadratic effect associated with down. There was also a positive effect associated with 5) "game seconds remaining," which implies that, overall, per carry yardage was greater earlier in the game. For 6) the "score differential," which is the difference between the possessing team's score and their opponent's score, the negative effect could be interpreted as: When the score is not in the favor of the offense, the rushing yards per carry are greater. Next, as 7) "yards to go" increased, so did rushing yards per carry. Relatedly, as the 8) "line of scrimmage" (recorded as yards to the opponent's endzone) increases, we also see evidence for greater yards per carry. To some extent, this is a function of the compression of the field that occurs when there are fewer yards to gain before a touchdown occurs. As we would expect, when the 10) "opponent's EPA" is large, they had a worse defense and thus allowed more rushing yards per carry.

Overall, these results are sensible and provide quantitative evidence for the aspects of player performance we hypothesized *a priori*. By contrast, we did not find evidence for effects of 2) "age squared," although we noted perhaps a mild negative effect that we would expect, 7) "days of rest" which was surprising given that on 697 carries in the dataset, the player only had 4 days of rest between games, whereas there were 646 instances where the player had at least 2 weeks of rest. However, it is important to remember that we modeled yards per carry, thus, it is possible that some players with less rest between games may have had longer but fewer runs, due to fatigue or injury. We did not find evidence for 9) a home field advantage, but again, we did see a mild effect and it is possible that this effect could vary by player because stadiums and fan bases vary in their noise level. Also, if anything, the home field may have more influence on quarterback performance and offensive line penalties, so we may not expect to see a strong effect on running back yardage. Finally, 10) "travel distance" to commute to the game from the home city did not seem to be a strong effect in either model, at least not when considering the other covariates simultaneously.

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As in the single-player analysis, we estimated a relatively small τ (95% CI: (0.27,0.29)), indicating a substantial right-skew to the data overall when considering the heterogeneous trend due to covariates. Thus, the ALD base distribution is important for characterizing the stochasticity associated with running back yardage across the NFL during the 2023 season.

4 Discussion

We highlighted the statistical implications of the unique way yardage is recorded in football. Rather than rounding the true yardage to the nearest whole number, recorded yardage is calculated as the difference between the rounded line of scrimmage (b) and the rounded yard line where the player is downed (d). Importantly, the form of rounding in the data occurs as a floor operation (i.e., rounded down), but our approach also holds for standard rounding of b and d (i.e., to the nearest integer) if that were the case.

We provided a general framework to model yards gained using a chosen base probability distribution. We demonstrated this method through both single- and multi-player analyses, focusing on running backs during the 2023 NFL season. Code to generate the rounded-normal and rounded-ALD distributions, as well as to reproduce these analyses, is available at https://github.com/akglazer/rounded-sports-data.

Our approach relied on flexible rounded distributions for the yardage data using a transformation of variables, and we demonstrated their application by fitting rounded-normal and rounded-ALD models to a single player's (Bijan Robinson) 2023 rushing yards. We found that the rounded-ALD model was superior to the rounded-normal in this setting. By considering the correct rounded model, we gained statistical efficiency and were able to detect a significant effect ("seconds remaining") that predicts true yardage. Furthermore, using the naive approach, a basic multiple linear regression model for the rounded data y, we were not able to detect that helpful predictor. It is well-known that accounting for rounding can alleviate various statistical issues that might arise otherwise, and we confirmed this in our analysis, even in the case of such unusual rounding in yardage data.

We extended this modeling approach to the multiplayer setting and specified a hierarchical rounded-ALD model using a random intercept for each player. Like the single-player model, the rounded-ALD multi-player model also resulted in an improved fit over the rounded-normal model. Accounting for the correctly rounded and asymmetric distribution allowed us to identify additional covariates that were predictive of rounding at the league level.

In recent years, the use of player-tracking has become standard practice not only in football but across a wide array of professional sports (e.g., basketball, soccer, and ice hockey; Kovalchik 2023). These tracking tools provide high-resolution, spatiotemporal coordinates of every athlete on the field/court, enabling a new wave of statistical and machine-learning methods for tasks such as quantifying player speed and acceleration, classifying defensive coverages or offensive route concepts, measuring defensive pressure, and analyzing fourth down strategies (e.g., Lopez, 2020; Nguyen et al., 2024; Dutta et al., 2020; Mallepalle et al., 2020)).

Although the NFL has occasionally released portions of its player-tracking data, for example, through the NFL Big Data Bowl competitions, those public releases have been limited to small subsets of games and are not maintained as open datasets. Indeed, the files posted for the 2024 and 2025 competitions are no longer available for download. Crucially for our

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application, the publicly released data describe only player movement, not the precise position of the football itself. While the league has indicated that a radio frequency identification (RFID) sensor is embedded in the ball, details on when and how the corresponding ball-tracking data are recorded, processed, or used are not clear. Even with tracking systems, measurement error can occur from sources such as blocking of sensors or players, limited sampling frequency, systematic error from sensor placement, and signal interference (e.g., Roell et al., 2019; Blauberger et al., 2021)). Comprehensive and publicly accessible ball-tracking data could be paired with conventional recorded data in a joint statistical model providing a higher resolution source of information about yardage, but such data are not released at the time of this writing. Furthermore, the analysis of historical data that pre-dates ball-tracking will still require an explicit treatment of rounding.

Beyond football, our findings underscore the importance of taking into account processes that introduce discretization or asymmetry, because inference can be meaningfully impacted otherwise. Our approach offers not only a strategy to account for the unique rounding procedure associated with officially recorded yardage data, but also a more general framework for analysts working with integer-valued data derived from continuous phenomena.

We acknowledge some limitations and directions for future work. First, we assumed that yardlines are always rounded down (i.e., toward the goal line); however, this is not always the case (National Football League, 2024). Consider the following scenario: The offense faces 3rd down with 1.4 yards to go from the opponent's 30.6-yard line. Under NFL record-keeping rules, this is recorded as 3rd and 1 from the 30-yard line. Suppose the offense then advances to the 29.6-yard line, which is a physical gain of one yard. However, the remaining yards to go (0.4) is forced to display as 1 yard rather than 0, the recorded ball position is still listed as the 30-yard line, and the official system results in no gain (or zero yards on that play). A one-yard physical advancement is effectively erased in the recorded play-by-play data. Plays recorded as 1 yard-to-go account for 4.7% of all NFL plays in the 2023 season. A subset of these plays, such as the preceding example, result in yardage gains that are not recorded according to the rounding rules described earlier. There are other unique situational record-keeping protocols in the NFL, such as when the line of scrimmage is less than 10 yards to the end zone, and these situations could be accommodated with additional modifications to the modeling framework.

Second, we focused on rushing plays, but extending this work to passing and receiving yards would provide a more complete picture of offensive performance. Unlike rushing plays, where the start and end of the play are typically well-defined in terms of field position, passing and receiving yards can involve more complex spatial dynamics, such as air yards versus yards after catch, and are often influenced by forward progress rulings and spot-of-catch locations. These aspects introduce additional sources of measurement error and rounding discrepancies, which may differ systematically from those affecting rushing plays. Moreover, the interaction among ball placement, receiver motion, and defensive positioning in passing plays may lead to different patterns of rounding bias, particularly near the sidelines or in high-leverage game situations. Capturing and correcting for such rounding effects in passing data would require refinements to the statistical framework we presented, but doing so would broaden the applicability and impact of this line of research.

Finally, the true yards $\tilde{y}$ are marginally bounded by -100 below and by 100 above because the length of the field in American football is 100 yards. However, conditionally, $\tilde{y}$ is

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constrained based on the line of scrimmage b such that $b-100<\tilde{y}<b$. This compact support for $\tilde{y}$ can be accounted for using a truncated distribution for the true yardage model $f_{\tilde{y}}(\tilde{y})$. However, recall that $\lfloor b \rfloor$ is recorded instead of the true line of scrimmage b. Thus, the unknown truncation needs to be accounted for in $f_{\tilde{y}}(\tilde{y})$. In Appendix A, we show how to derive a truncated distribution for yardage data that accounts for the unknown truncation bounds. In our case, when applying the truncated model, the results were not meaningfully different from those we reported because there were relatively few plays close enough to the opponent's endzone that the truncation changed the form of the distribution.

As the field of sports analytics continues to evolve, careful attention to how data are generated and recorded is essential to ensure valid conclusions. Ideally, the underlying continuous data would be accurately recorded and publicly available. However, in the absence of such data, analysts must be thoughtful about how recorded values are treated if a generative statistical model is desired. Overlooking these details, such as the unique rounding mechanism used to record yardage in football, can lead to misleading inferences.

Disclosure statement

The authors have no conflicts of interest to declare.

Data Availability Statement

Data are available at the following URL: https://github.com/akglazer/rounded-sports-data.

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Appendix A: Truncated Yardage Distributions

True yardage $\tilde{y}$ is bounded in a way that depends on the current line of scrimmage *b* such that $b-100<\tilde{y}<b$; however, only $\lfloor b \rfloor$ is recorded and thus $b \approx \lfloor b \rfloor + u$ for $u \sim \text{Unif}(0,1)$. For an untruncated continuous random variable $\dot{y}$ with distribution $f_{\dot{y}}(\dot{y})$, the associated conditional truncated distribution for $\tilde{y}$ is

$$f_{\tilde{y}}(\tilde{y}\,|\,u) = \frac{f_{\dot{y}}(\tilde{y})}{F_{\dot{y}}(b) - F_{\dot{y}}(b-100)} \quad (31)$$

$$= \frac{f_{\dot{y}}(\tilde{y})}{F_{\dot{y}}(\lfloor b \rfloor + u) - F_{\dot{y}}(\lfloor b \rfloor - 100 + u)}. \quad (32)$$

Thus, because *u* is unknown, we can marginalize over *u* to obtain the truncated distribution

$$\bar{f}_{\tilde{y}}(\tilde{y}) = \int_0^1 f_{\tilde{y}}(\tilde{y}\,|\,u)\, f_u(u)du\,. \quad (33)$$

To compute the data model $p_y(y)$ for recorded yardage *y* in (16), we need the CDF associated with (33). To obtain the CDF $\bar{F}_{\tilde{y}}(\tilde{y})$ based on a general continuous untruncated distribution $f_{\dot{y}}$, we derive an integral form that can approximated using quadrature as

$$\bar{F}_{\tilde{y}}(\tilde{y}) = \int_{-\infty}^{\tilde{y}} \bar{f}_{\tilde{y}}(v)dv \quad (34)$$

$$= \int_{-\infty}^{\tilde{y}} \int_0^1 f_{\tilde{y}}(v\,|\,u)\, f_u(u)dudv \quad (35)$$

$$= \int_0^1 \int_{-\infty}^{\tilde{y}} f_{\tilde{y}}(v\,|\,u)dv f_u(u)du \quad (36)$$

$$= \int_0^1 F_{\tilde{y}}(\tilde{y}\,|\,u)\, f_u(u)du \quad (37)$$

$$= \int_{-1}^1 F_{\tilde{y}}\left(\tilde{y}\,|\,\frac{\phi}{2}+\frac{1}{2}\right) f_\phi(\phi)d\phi\,, \quad (38)$$

where we used the change of variables $u = \frac{\phi}{2} + \frac{1}{2}$ for $f_\phi(\phi) = \text{Unif}(-1,1)$. The form in (38) can be approximated using Gauss-Legendre quadrature, although it requires a large number of quadrature points which can reduce efficiency of an algorithm that relies on numerous evaluations of (38).

We can use (38) to compute the PMF for the rounded data *y*, using (16) as

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$$p_y(y) = \int_{-1}^{1} \left(\bar{F}_{\tilde{y}}\left(y + \frac{\theta}{2} + \frac{1}{2} \right) - \bar{F}_{\tilde{y}}\left(y + \frac{\theta}{2} - \frac{1}{2} \right) \right) f_\theta(\theta) d\theta \,, \quad (39)$$

where $f_\theta(\theta) = \text{Unif}(-1,1)$ as in (16). This final PMF for rounded truncated yardage y requires multiple quadrature approximations and can be computationally intensive to evaluate for large datasets. However, we applied (39) to analyze the single player data from Bijan Robinson in Section 2 and found the results were not meaningfully different using the truncated distribution in (39) (Figure 5).

We attribute that to relatively few plays ($\approx 5\%$) near enough to the endzone that the form of distribution was affected by the truncation. The negligible change in inference combined with the computational limitations in using the truncated data model in (39) for large datasets implies that it may not be justified unless a significant improvement in computational efficiency is found.

Notes

[1] Despite the static yardage markers that are painted on each side of the field decreasing as they approach the end zones from the 50-yard line, the data are structured such that the position on the field is from the offensive perspective. Thus, for the offensive ball carrier, $b_{ij} = 75$ indicates that there are 75 yards to reach the opponent's end zone.

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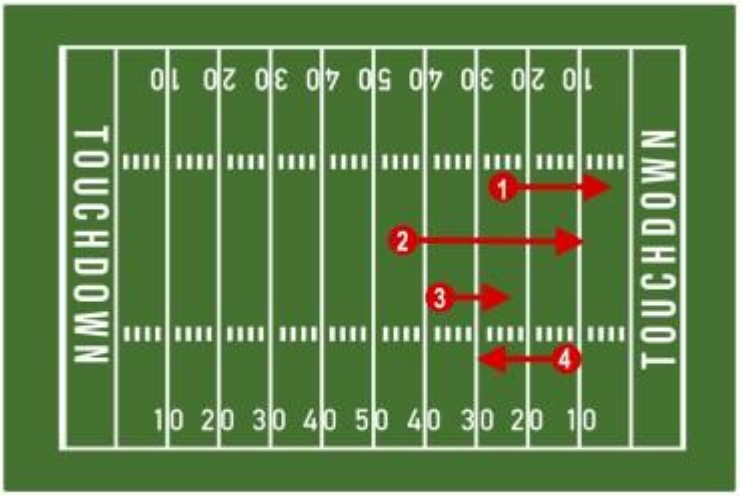
Rushing Plays



Truth

Play	Carry began, b_j	Player downed, d_j	Yards gained, $\tilde{y}_j$
j = 1	25.2	4.6	20.6
j = 2	44.7	10.1	34.6
j = 3	38.4	24.6	13.8
j = 4	12.3	29.9	-17.6

Recorded

Play	Carry began, $\lfloor b_j \rfloor$	Player downed, $\lfloor d_j \rfloor$	Yards gained, y_j
j = 1	25	4	21
j = 2	44	10	34
j = 3	38	24	14
j = 4	12	29	-17

Figure 1: Illustration of rounding in recorded yardage using four example plays, $j = 1, 2, 3, 4$; b_j and d_j are always in terms of yards to go to the opponent's endzone. In this figure, the opponent's endzone is the endzone on the right.

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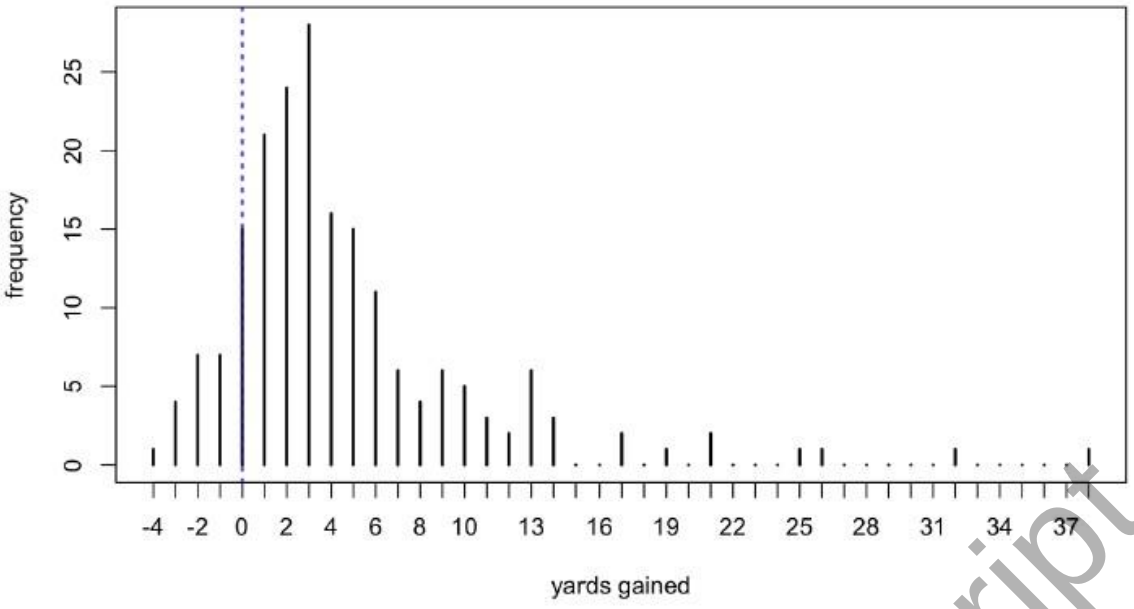

Figure 2: Recorded yardage data for Bijan Robinson based on $J = 193$ carries in 2023. Dashed vertical line demarcates zero.

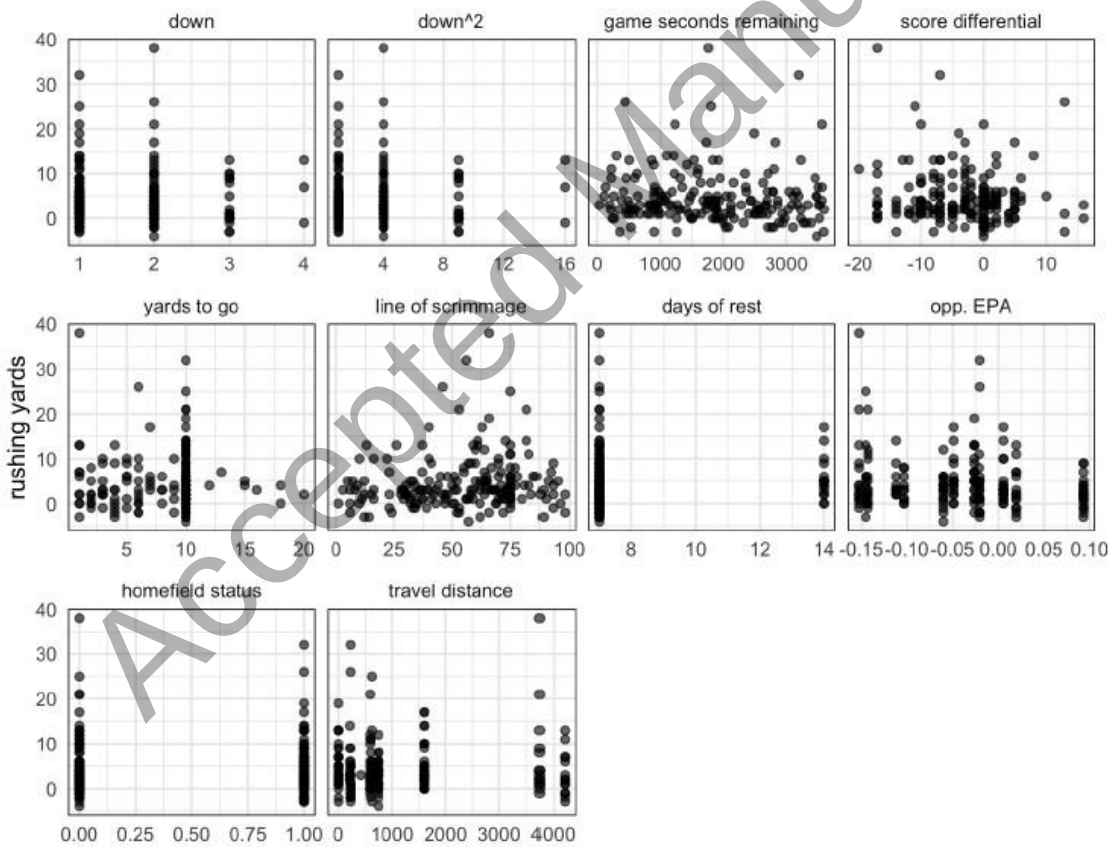

Figure 3: Recorded yardage data for Bijan Robinson versus covariates.

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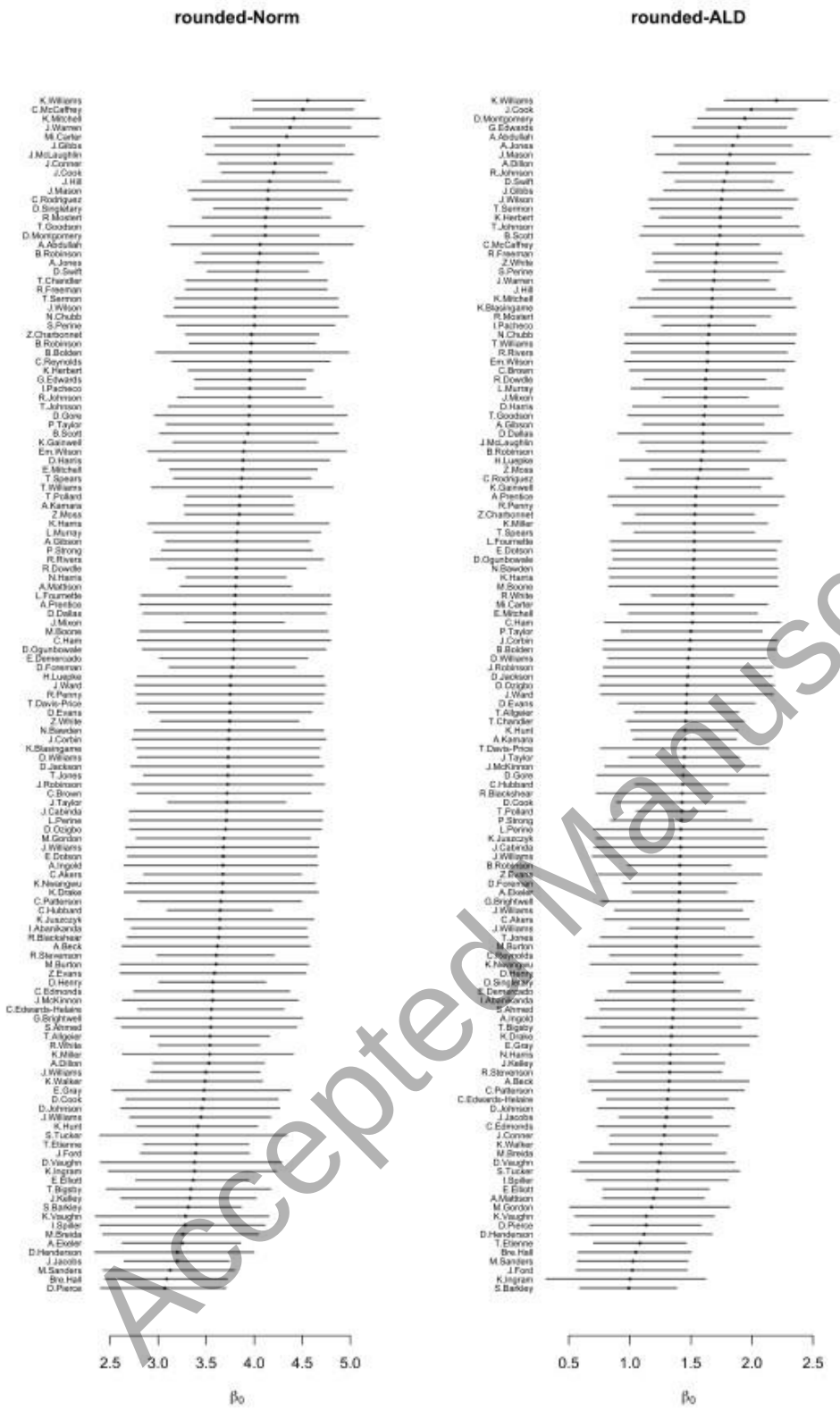

Figure 4: Posterior mean and 95% equal-tailed credible intervals for $\beta_{0,i}$ (random effect intercepts for players) resulting from the fit of each model to the multi-player data.

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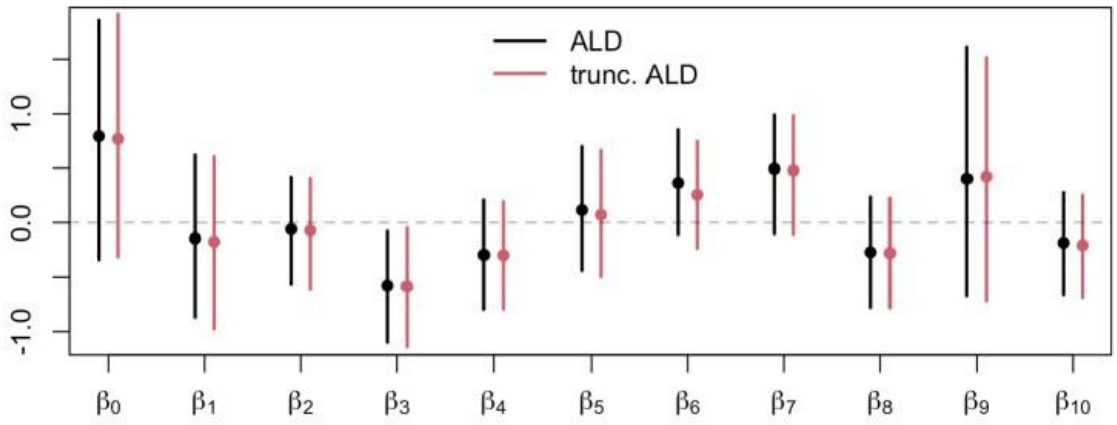

Figure 5: Posterior mean and 95% credible intervals for each coefficient (β_j) for the single player analysis based on the rounded ALD model (black) and the rounded truncated ALD model (red).

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Table 1: Posterior characteristics for $\boldsymbol{\beta}$ from both candidate models.

		rounded-Normal			rounded-ALD		
	covariate	mean	sd	95% CI	mean	sd	95% CI
$\boldsymbol{\beta}_1$	down	0.03	0.73	(-1.39, 1.49)	-0.14	0.38	(-0.87, 0.62)
$\boldsymbol{\beta}_2$	down squared	0.06	0.39	(-0.72, 0.81)	-0.06	0.25	(-0.57, 0.42)
$\boldsymbol{\beta}_3$	seconds remaining	-0.32	0.44	(-1.16, 0.55)	-0.58	0.26	(-1.10, -0.08)
$\boldsymbol{\beta}_4$	score diff	-0.56	0.49	(-1.51, 0.41)	-0.30	0.26	(-0.79, 0.21)
$\boldsymbol{\beta}_5$	yards to go	-0.34	0.58	(-1.47, 0.81)	0.12	0.29	(-0.44 0.70)
$\boldsymbol{\beta}_6$	line of scrimmage	0.47	0.49	(-0.15, 1.68)	0.37	0.24	(-0.11, 0.85)
$\boldsymbol{\beta}_7$	days of rest	0.32	0.47	(-0.60, 1.26)	0.49	0.28	(-0.11, 0.99)
$\boldsymbol{\beta}_8$	opponent's rush EPA	-0.73	0.48	(-1.70, 0.20)	-0.27	0.26	(-0.78, 0.24)
$\boldsymbol{\beta}_9$	homefield	1.16	1.00	(-0.71, 3.14)	0.40	0.58	(-0.68, 1.61)
$\boldsymbol{\beta}_{10}$	travel distance	0.10	0.45	(-0.75, 1.01)	-0.18	0.24	(-0.67, 0.28)

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Table 2: Posterior characteristics for $\boldsymbol{\beta}$ from both candidate multi-player models.

		rounded-Normal			rounded-ALD		
	covariate	mean	sd	95% CI	mean	sd	95% CI
$\boldsymbol{\beta}_1$	age	-0.02	0.08	(-0.17, 0.13)	0.11	0.05	(0.01, 0.21)
$\boldsymbol{\beta}_2$	age^2	0.02	0.05	(-0.07, 0.11)	-0.04	0.03	(-0.10, 0.02)
$\boldsymbol{\beta}_3$	down	0.23	0.07	(0.09, 0.37)	0.12	0.12	(0.03, 0.21)
$\boldsymbol{\beta}_4$	down^2	-0.05	0.04	(-0.13, 0.03)	-0.05	0.03	(-0.10, 0.00)
$\boldsymbol{\beta}_5$	seconds remaining	0.03	0.05	(-0.06, 0.12)	0.07	0.03	(0.01, 0.13)
$\boldsymbol{\beta}_6$	score diff	-0.10	0.05	(-0.20, 0.00)	-0.08	0.03	(-0.14, -0.01)
$\boldsymbol{\beta}_7$	yards to go	0.45	0.05	(0.33, 0.56)	0.31	0.03	(0.23, 0.38)
$\boldsymbol{\beta}_8$	line of scrimmage	0.34	0.06	(0.25, 0.44)	0.21	0.03	(0.15, 0.27)
$\boldsymbol{\beta}_9$	days of rest	0.05	0.05	(-0.04, 0.14)	0.01	0.03	(-0.06, 0.07)
$\boldsymbol{\beta}_{10}$	opponent's rush EPA	0.10	0.05	(0.01, 0.19)	0.11	0.03	(0.05, 0.17)
$\boldsymbol{\beta}_{11}$	homefield	0.20	0.09	(0.02, 0.38)	0.07	0.06	(-0.06, 0.20)
$\boldsymbol{\beta}_{12}$	travel distance	0.02	0.05	(-0.07, 0.11)	0.04	0.03	(-0.03, 0.10)

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